

Department of Computer Science
Gujarat University
MCA sem2
Assignment for unit 3,4,5

1. Define Knowledge Representation and its importance in AI.
2. What is a Semantic Network? Give one example.
3. Differentiate between: Propositional Logic vs Predicate Logic
4. What is Conceptual Dependency?
5. Define Forward and Backward Chaining.
6. What are Predicates and Functions in logic?
7. Explain Procedural vs Declarative Knowledge.
8. List the steps of NLP.
9. What is a Perceptron?
10. Define Expert System with one real-life example.
11. Explain Knowledge Representation Techniques:
 - Semantic Networks
 - Partitioned Semantic Networks
 - Conceptual Dependency
 - Rule-based systems
12. Discuss advantages and limitations of Knowledge Representation.
13. Explain First Order Predicate Logic (FOPL) with examples.
14. Compare: Forward Chaining vs Backward Chaining (with diagrams and examples)
15. Explain Minimax Algorithm with an example.
16. Describe Alpha-Beta Pruning and how it improves efficiency.
17. Explain Backpropagation Algorithm in Neural Networks.
18. Discuss Expert Systems:
 - Components
 - Characteristics
 - Advantages & Disadvantages
19. Write Predicates of below sentences:
 - All humans are mortal.
 - Socrates is a human.
 - Therefore, Socrates is mortal.
 - Every student studies AI.
 - Some animals are mammals.
 - No bird is a mammal.
20. Write logical expression related to:
 - "Every teacher teaches some subject."
 - "There exists a student who likes AI."
21. Draw a semantic network for:
 - Animal $\rightarrow$ Bird $\rightarrow$ Sparrow
 - Bird can fly
 - Sparrow is a bird
22. Explain inheritance property in semantic networks.
23. What are ISA links and HAS-A relationships?
24. Explain:
 - Tokenization

Parsing

Semantic Analysis

25. List real-world NLP applications.

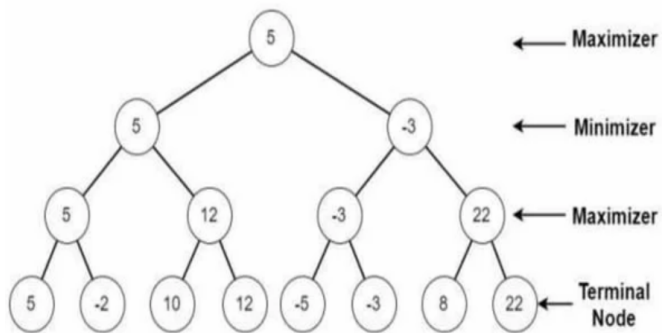
26. Difference between: Biological vs Artificial Neural Networks

27. Explain:

Single Layer Perceptron (SLP)

Multi-Layer Perceptron (MLP)

28. Apply Minimax Algorithm on a given tree.



29. Show how Alpha-Beta pruning reduces nodes in above tree.

30. Write a short note on expert system Mycin.